NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83401/18A(USAF)
9 October 1989
SUPERSEDING
MIL-R-83401/18(USAF)
9 June 1986

PERFORMANCE SPECIFICATION

RESISTOR NETWORK, FIXED, FILM,
STYLE RZ180, 6 PIN SIP

This specification is approved for use within Electronic support Division AFLC, Department of the AIR Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RZ180, 6 pin single-in-line package (SIP) resistor networks. This style is available in all characteristic K and resistance tolerances G.

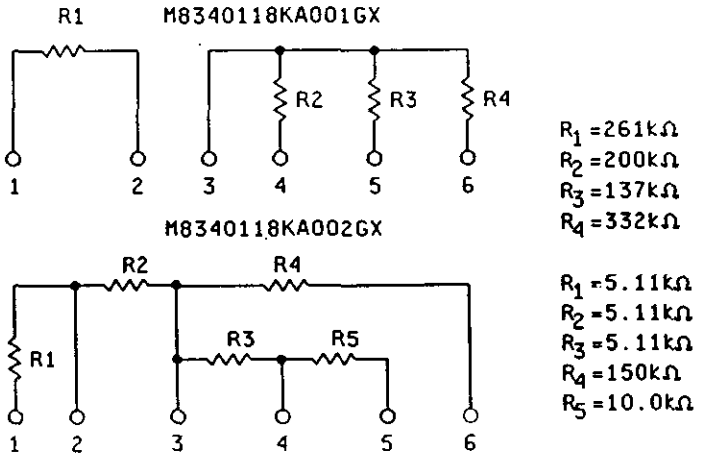
1.2 Military part number. *Resistor networks covered by this specification* shall be identified by a military part number which shall consist of the basic number of this specification and a coded number. The military part number shall be in the following form:

M8340118 — Detail specification number

KA001GX — Coded number

The coded number shall be derived in accordance with MIL-R-83401.

1.2.1 Schematics. *The schematic of the resistor network* shall be identified by the following part number in accordance with the drawings below.



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5905

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:58Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-39032 - Resistors, Packaging of.
MIL-R-83401 - Resistor Networks, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Resistors, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-83401, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating (70°C). The power ratings shall be as follows:

Part number	Element power rating (watt)	Network Power rating (watt)
M8340118KA001GX	0.10	0.40
M8340118KA002GX	0.10	0.50

3.4 Power conditioning. The power applied for power conditioning shall be 1.5 times rated power.

3.5 Voltage rating. The maximum continuous working voltage for each resistor shall not exceed 50 volts dc or ac rms.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-83401.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:58Z
Check the source to verify that this is the current version before use.

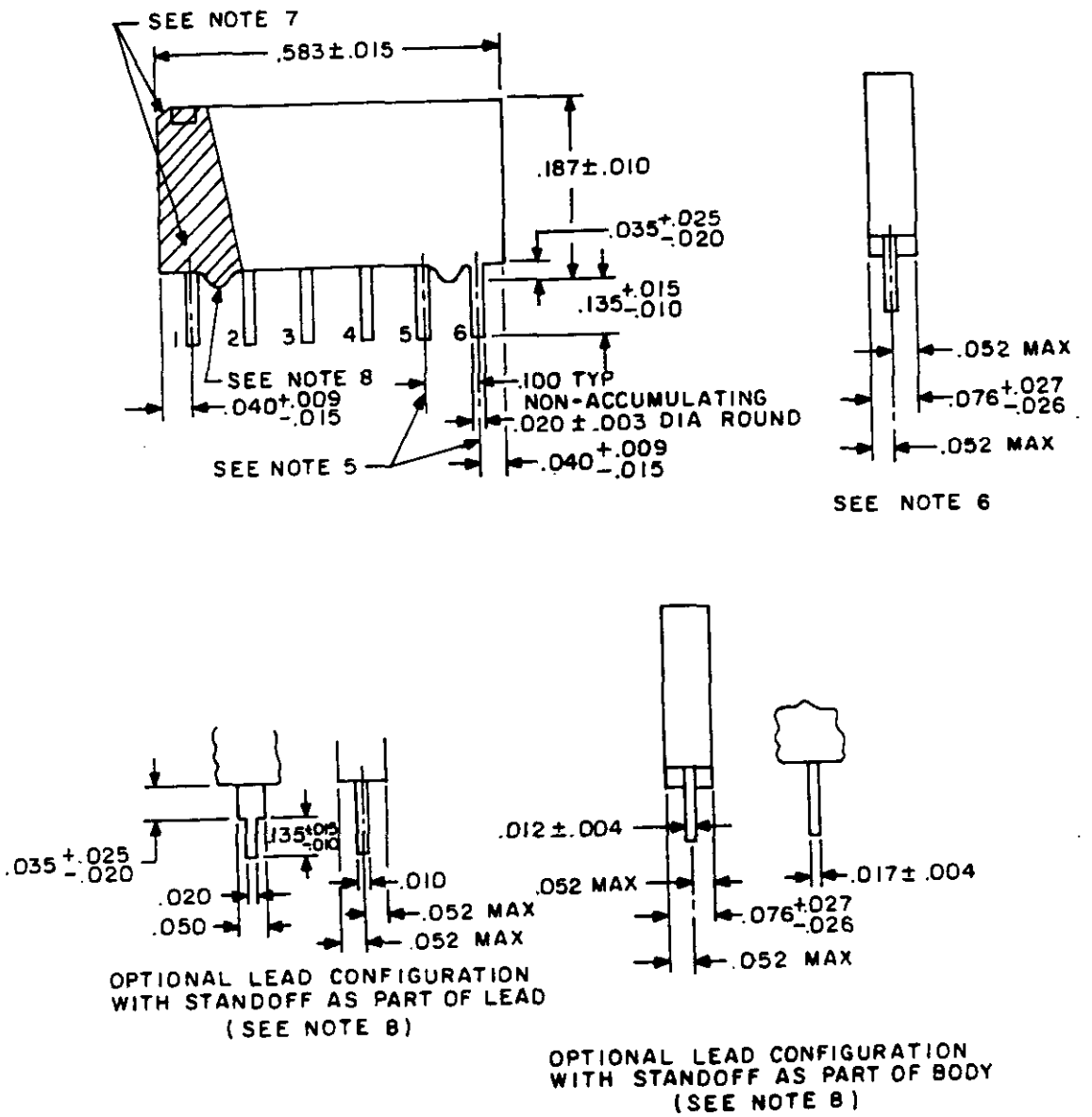


FIGURE 1. Style RZ180 resistors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:58Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm
.003	0.08	.025	0.64
.004	0.10	.030	0.76
.005	0.13	.035	0.89
.008	0.20	.040	1.02
.009	0.23	.049	1.24
.010	0.25	.050	1.27
.012	0.30	.074	1.88
.014	0.36	.089	2.26
.015	0.38	.100	2.54
.017	0.43	.135	3.43
.020	0.51	.187	4.75
.024	0.61	.583	14.81

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. The picturization of the styles above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device, are acceptable.
5. Terminal centerline to centerline measurements made at point of emergence of the lead from the body.
6. Measurement made at the point of emergence of the lead from the body.
7. Pin 1 locator shall be a dot, notch, stripe, or numeral 1 adjacent to pin no. 1, in the shaded area; also, additional marking may be placed on the top edge where the bevel may be located.
8. If the standoffs are located on the body, a minimum of two standoffs are required as illustrated. As an option, additional standoffs may be located on the body of the resistor network. If leads with standoffs are used, standoffs on the body are not required.

FIGURE 1. Style RZ180 resistors - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:58Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-83401 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Extension of qualification. Qualification to schematic B of style RZ070 will qualify style RZ180.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5905-F297)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:58Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-83401/18A(USAF)	Resistor Network, Fixed, Film, Style RZ180, 6 Pin SIP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-05-23T07:58Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE
2750 ABW/ES
Gentile AF Station
Dayton, OH 45444

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

Electronic Support Division AFLC
2750 ABW/ES
Gentile AF Station
Dayton, OH 45444

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:58Z
Check the source to verify that this is the current version before use.